

Report 5

Your Name

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1 Update K setting in CVX

1.1 Stabilizing Sparse Feedback Gains via LMI Repair

In particular, for a discrete-time linear system $x_{t+1} = (A + BK)x_t$, we formulated an SDP to solve for a positive-definite matrix P and feedback matrix K such that the Lyapunov inequality

$$(A + BK)^\top P (A + BK) - P \prec -\epsilon I$$

is satisfied, with sparsity enforced on K via masked indexing.

In our MATLAB implementation, we created a script that exhaustively scans over feasible combinations of `cts` (number of active rows and columns) for a fixed `bin`, and records which configurations admit a stabilizing sparse controller.

This LMI-repair module was integrated as a fallback step in our evolutionary search framework: after sparsity constraints are applied and instability is detected (e.g., infinite LQR cost), we attempt a final repair via convex optimization before discarding the sample(to be done).

Currently, it can not be solved properly because it's bilinear and non convex.

2 Strategy: Discovering Sparse Controllers via Evolving LQR Weighting Matrices

2.1 System and Objective Formulation

$$\vec{x}_{t+1} = A\vec{x}_t + B\vec{u}_t \tag{1}$$

$$\vec{x}_{t+1} = (A + BK)\vec{x}_t \tag{2}$$

$$J = \sum_{t=0}^{\infty} (\vec{x}_t^T Q \vec{x}_t + \vec{u}_t^T R \vec{u}_t) \quad (3)$$

2.2 Evolutionary Co-Design Strategy

Chromosome Representation:

$$\vec{\theta} = [\text{diag}(Q)^T, \text{diag}(R)^T]^T \in \mathbb{R}^{n+m} \quad (4)$$

Gene Expression Process (Chromosome to Controller):

1. **Construct Weighting Matrices:** Build the diagonal matrices $Q = \text{diag}(\vec{\theta}_{1:n})$ and $R = \text{diag}(\vec{\theta}_{n+1:n+m})$.
2. **Solve for the Optimal Dense Controller:** Solve the Discrete Algebraic Riccati Equation (DARE) to find the unique positive semi-definite solution P . Then, compute the optimal dense controller K_{dense} , which is guaranteed to stabilize the system.

$$K_{\text{dense}} = (R + B^T P B)^{-1} (B^T P A) \quad (5)$$

3. **Sparsification:** Apply a sparsification operator $\mathcal{S}_k(M)$, which retains the k elements of matrix M with the largest magnitudes and sets all others to zero. This yields the sparse controller:

$$K_{\text{sparse}} = \mathcal{S}_{k_{\text{max}}}(K_{\text{dense}}) \quad (6)$$

Fitness Function:

$$J_{\text{total}}(\vec{\theta}) = \begin{cases} J_{\text{LQR}}(K_{\text{sparse}}) + C_{\text{hardware}}, & \text{if } \rho(A - BK_{\text{sparse}}) < 1 \\ J_{\text{penalty}}, & \text{otherwise} \end{cases} \quad (7)$$

3 Performance of EA

3.1 LQR Cost Optimization via Evolution of Q and R

To explore an alternative co-design strategy, we replaced the direct optimization of the feedback gain matrix K with evolutionary updates to the cost matrices Q and R in the discrete-time LQR formulation. Given that the optimal K is computed via the discrete algebraic Riccati equation (DARE), this approach enables indirect control over the feedback policy while retaining closed-form stability guarantees when the Riccati solution exists.

In this framework, individuals in the population encode flattened versions of symmetric matrices Q and R , constrained to be positive semidefinite and of fixed dimensions. After reshaping and projecting to the positive semidefinite cone, each candidate is used to compute the corresponding LQR gain K via MATLAB's `dlqr()` function. The resulting cost $\text{tr}(P)$ is normalized against the benchmark LQR cost computed from identity weights and serves as the fitness metric.

As shown in Fig. 1 and Fig. 2, the evolutionary process leads to monotonic improvements in both the best and average costs per generation. Additionally, Fig. 3 confirms that all individuals remain stabilizing throughout the search, benefiting from the Riccati-based construction.

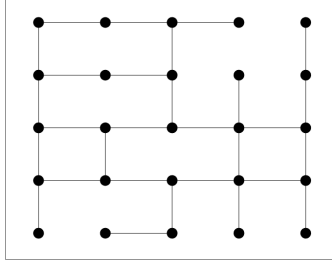


Figure 1: Best normalized LQR cost per generation with evolving Q, R .

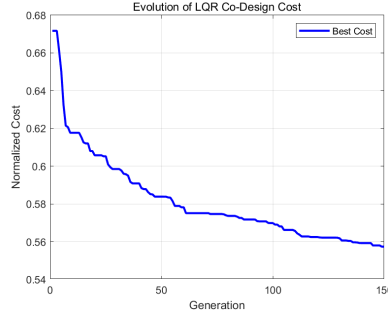


Figure 2: Average normalized LQR cost per generation.

Define an *emergence score*

$$\text{Em}(n, \alpha) := \max \left\{ 0, -\frac{1}{n} \log \hat{\beta}_{\alpha}^{\star}(n) \right\} \approx \hat{D}(\mathcal{P}_0 \| \mathcal{P}_1) - \frac{c_{\alpha, n}}{n},$$

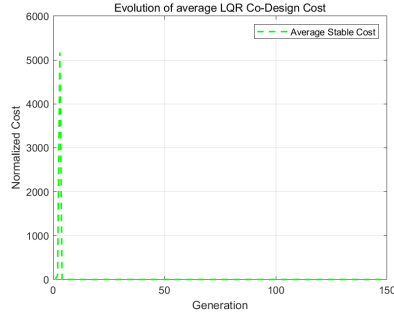


Figure 3: Number of unstable individuals per generation (all zero).

and report $(p\text{-value}, \mathbf{Em})$ with CIs. Larger $\mathbf{Em}$ indicates stronger statistical evidence for a structural departure.

4 What to Report (ML topics)

- Extract evry col or row(or there features) as an attribute. Use decision tree and RF to learn some empirical ideas about the distribution of the possible positions that the matrix can be stable and have low cost.
- $\hat{D}(\mathcal{P}_0\|\mathcal{P}_1)$, finite-sample upper bound on $\beta_\alpha^\star(n)$, setting the divergence versus the best pop as the optimal objective.